Anytime Moving-Center DDLALM for Nonsmooth Composite Optimization

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Research update, August 2026

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I. Problem Setting

Structured Nonsmooth Composite Optimization

We consider

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \quad F(\boldsymbol{x}) := f(\boldsymbol{x}) + h\big(\mathcal{A}(\boldsymbol{x})\big). \tag{P0}$$

- $f : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable and may be nonconvex.
- $\mathcal{A} : \mathbb{R}^n \to \mathbb{R}^m$ is continuously differentiable and may be nonlinear.
- $h : \mathbb{R}^m \to \mathbb{R}$ is convex, Lipschitz continuous, and possibly nonsmooth.

Approximate KKT Point

- Introduce $\boldsymbol{y} = \mathcal{A}(\boldsymbol{x})$:

$$\min_{\boldsymbol{x},\boldsymbol{y}} \quad f(\boldsymbol{x}) + h(\boldsymbol{y}) \qquad \text{s.t.} \qquad \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y} = \boldsymbol{0}. \tag{P}$$

- The split KKT system is

$$\nabla f(\boldsymbol{x}) + J_{\mathcal{A}}(\boldsymbol{x})^{\top}\boldsymbol{q} = \boldsymbol{0}, \qquad \boldsymbol{q} \in \partial h(\boldsymbol{y}), \qquad \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y} = \boldsymbol{0}.$$

- **approximate ε-KKT point.** Find $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{q})$ such that

$$\|\nabla f(\boldsymbol{x}) + J_{\mathcal{A}}(\boldsymbol{x})^{\top}\boldsymbol{q}\| \le \varepsilon, \quad \boldsymbol{q} \in \partial h(\boldsymbol{y}), \quad \|\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\| \le \varepsilon.$$

  We denote the maximum of these three residuals by $\mathcal{R}_{\text{split}}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{q})$.
- The exact $\boldsymbol{y}$-proximal step makes the subdifferential condition exact.

Main Result

The Candidate method uses the four-block order

$$\boxed{\boldsymbol{x} \longrightarrow \boldsymbol{\lambda} \longrightarrow \boldsymbol{z} \longrightarrow \boldsymbol{\mu}}$$

and the anytime schedule

$$r_k := s_k^{-1} = \Theta(k^{-1/3}), \qquad \beta_k = \Theta(k^{-1/3}), \qquad \gamma_k = \Theta(k^{-1}).$$

Under the stated smoothness, lower-boundedness, and global Lipschitz assumptions, one tail iterate satisfies

$$\mathcal{R}_{\mathrm{split}}(\boldsymbol{x}^{\kappa+1}, \boldsymbol{y}_\kappa^{\mathrm{out}}, \boldsymbol{q}_\kappa^{\mathrm{out}}) = O(K^{-1/3}).$$

Candidate guarantee

Single loop, no algorithmic projection, no inner solve, fixed base penalty ρ, two $\mathrm{prox}_{h/\rho}$ evaluations per iteration, and $O(\varepsilon^{-3})$ split-KKT complexity.

Research Problems

Q1: Fixed penalty, if single-loop ? Q1: For single-loop first-order algorithms, if ρ is fixed, to acclerate the rate?
Q2: Can combinng the dual and smoothin improves the iterative complexity?

II. Reduced-Dual Reformulation

Smoothed Augmented Lagrangian

The smoothed augmented Lagrangian used in our method is

$$\mathcal{P}(x, y, \lambda; z, \mu) := \mathcal{L}_\rho(x, y, \lambda) + \frac{r}{2}\|x - z\|^2 - \frac{1}{2s}\|\lambda - \mu\|^2.$$

Value Function

1. From original problem to minmax problem; 2. Froem The value function

- **Eliminate the y-block through the ρ-prox:**

$$d_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{\mu}) := \min_{\boldsymbol{y}} \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\lambda}; \boldsymbol{\mu}), \qquad \boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x}) = \operatorname{prox}_{h/\rho}\left(\mathcal{A}(\boldsymbol{x}) + \frac{\boldsymbol{\lambda}}{\rho}\right).$$

- **Obtain an exact subdifferential certificate:**

$$\boldsymbol{\lambda} + \rho\big[\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x})\big] \in \partial h\big(\boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x})\big).$$

- **Apply Danskin's theorem:**

▸ Danskin derivation →

$$\nabla_{\boldsymbol{x}} d_s = \nabla f(\boldsymbol{x}) + J_{\mathcal{A}}(\boldsymbol{x})^\top \left\{\boldsymbol{\lambda} + \rho\big[\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x})\big]\right\},$$

$$\nabla_{\boldsymbol{\lambda}} d_s = \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x}) - \frac{\boldsymbol{\lambda} - \boldsymbol{\mu}}{s}, \qquad \nabla_{\boldsymbol{\mu}} d_s = \frac{\boldsymbol{\lambda} - \boldsymbol{\mu}}{s}.$$

Under the strongly convex and strongly concave, it holds that:

$$\boxed{\max_{\lambda} d(\lambda, z, \mu) = p(z, \mu) = \min_{x} \max_{\lambda} \mathcal{F}(x, \lambda; z, \mu)}$$

III. Candidate Algorithm

Candidate Algorithm: Four-Block Single Loop

Set $r_k := 1/s_k$ and use only the fixed-parameter mapping

$$\boldsymbol{y}(\boldsymbol{\lambda};\boldsymbol{x}) := \operatorname{prox}_{h/\rho}\left(\mathcal{A}(\boldsymbol{x}) + \frac{\boldsymbol{\lambda}}{\rho}\right), \qquad \boldsymbol{q}(\boldsymbol{\lambda};\boldsymbol{x}) := \boldsymbol{\lambda} + \rho\big[\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{\lambda};\boldsymbol{x})\big].$$

Candidate notation: $\boldsymbol{q}$ is the split-KKT multiplier and $\boldsymbol{z}$ is the primal center.
Given $(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{z}^k, \boldsymbol{\mu}^k)$, perform

$$\begin{aligned}
\boldsymbol{y}^k &= \boldsymbol{y}(\boldsymbol{\lambda}^k;\boldsymbol{x}^k), \qquad \boldsymbol{q}^k = \boldsymbol{q}(\boldsymbol{\lambda}^k;\boldsymbol{x}^k),\\
\boldsymbol{x}^{k+1} &= \boldsymbol{x}^k - c\left[\nabla f(\boldsymbol{x}^k) + J_{\mathcal{A}}(\boldsymbol{x}^k)^\top \boldsymbol{q}^k + r_x(\boldsymbol{x}^k - \boldsymbol{z}^k)\right],\\
\widehat{\boldsymbol{y}}^{k+1} &= \boldsymbol{y}(\boldsymbol{\lambda}^k;\boldsymbol{x}^{k+1}),\\
\boldsymbol{\lambda}^{k+1} &= \boldsymbol{\lambda}^k + \alpha\left[\mathcal{A}(\boldsymbol{x}^{k+1}) - \widehat{\boldsymbol{y}}^{k+1} - r_k(\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k)\right].
\end{aligned}$$

Interpretation

The first two blocks are one primal descent step and one explicit reduced-dual gradient ascent step.

Candidate Algorithm: Center Updates and Cost

Complete the iteration in the order

$$\boxed{\boldsymbol{x} \longrightarrow \boldsymbol{\lambda} \longrightarrow \boldsymbol{z} \longrightarrow \boldsymbol{\mu}}$$

with

$$\boldsymbol{z}^{k+1} = \boldsymbol{z}^k + \beta_k(\boldsymbol{x}^{k+1} - \boldsymbol{z}^k),$$
$$\boldsymbol{\mu}^{k+1} = \boldsymbol{\mu}^k + \gamma_k(\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^k).$$

Use the anytime schedules

$$r_k = c_r(k + k_0)^{-1/3}, \qquad s_k = r_k^{-1},$$
$$\beta_k = \frac{\theta_z}{\ell_\rho} r_k, \qquad \gamma_k = \frac{\theta_\mu}{k + k_0}, \qquad c, \alpha = \Theta(1/\ell_\rho).$$

Here $k_0 \geq 1$, $c_r > 0$, $0 < \theta_\mu \leq 1$, and $\theta_z > 0$.

Algorithmic properties

No projection, no inner loop, and two evaluations of the same $\mathrm{prox}_{h/\rho}$ mapping per iteration.

IV. Complexity Result

Candidate Assumptions and Step-Size Conditions

(i) h is proper, closed, convex, finite-valued, and globally L_h-Lipschitz.

(ii) f and $\mathcal{A}$ are continuously differentiable; ∇f and $J_{\mathcal{A}}$ are globally Lipschitz, and $\|J_{\mathcal{A}}(\boldsymbol{x})\| \le G_{\mathcal{A}}$.

(iii) $F_{\inf} := \inf_{\boldsymbol{x}}\{f(\boldsymbol{x}) + h(\mathcal{A}(\boldsymbol{x}))\} > -\infty$.

Define

$$a_\rho := L_f + L_{\mathcal{A}} L_h + \rho G_{\mathcal{A}}^2, \qquad \ell_\rho := a_\rho + 2G_{\mathcal{A}} + \rho^{-1}.$$

Choose $r_x > 3\ell_\rho$, $r_0 \le \ell_\rho$, and fixed $c, \alpha > 0$ such that

$$c < \frac{1}{r_x + \ell_\rho + r_0},$$

$$\alpha \le \min\left\{ \frac{1}{11(\ell_\rho + r_0)}, \frac{c^2 m_0^2}{4(\ell_\rho + r_0)(1 + cm_0)^2}, \frac{1}{\rho^{-1} + r_0} \right\}, \quad m_0 := r_x - \ell_\rho - r_0.$$

Choose $\theta_z > 0$ sufficiently small so that every β_k satisfies all three caps stated in Lemma C4.2. This choice is uniform in k, K, and ε; moreover $0 < \gamma_k \le 1$ follows from $k_0 \ge 1$ and $0 < \theta_\mu \le 1$.

Candidate Theorem: $O(\varepsilon^{-3})$ Split-KKT Complexity

For each k, define the post-primal output

$$\boldsymbol{y}_k^{\mathrm{out}} := \widehat{\boldsymbol{y}}^{k+1}, \qquad \boldsymbol{q}_k^{\mathrm{out}} := \boldsymbol{\lambda}^k + \rho\big[\mathcal{A}(\boldsymbol{x}^{k+1}) - \boldsymbol{y}_k^{\mathrm{out}}\big].$$

Under the preceding assumptions and parameter conditions, for every $K \geq 4$ there exists one index

$$\kappa \in \{\lceil K/2 \rceil, \ldots, K-1\}$$

such that

$$\boldsymbol{q}_\kappa^{\mathrm{out}} \in \partial h(\boldsymbol{y}_\kappa^{\mathrm{out}}),$$

$$\left\| \nabla f(\boldsymbol{x}^{\kappa+1}) + J_{\mathcal{A}}(\boldsymbol{x}^{\kappa+1})^\top \boldsymbol{q}_\kappa^{\mathrm{out}} \right\| \leq C_{\mathrm{stat}} K^{-1/3},$$

$$\left\| \mathcal{A}(\boldsymbol{x}^{\kappa+1}) - \boldsymbol{y}_\kappa^{\mathrm{out}} \right\| \leq C_{\mathrm{feas}} K^{-1/3}.$$

Complexity

$K = O(\varepsilon^{-3})$ iterations and $2O(\varepsilon^{-3})$ fixed-parameter h-prox evaluations; the result is tail-best-iterate.

V. Proof Sketch

Proof Sketch I: Reduced Game and Frozen Descent

- **Module 1: reduced geometry and bounded dual orbit.** Eliminating $\boldsymbol{y}$ gives a smooth nonconvex-concave payoff and an exact split-KKT multiplier:

$$\nabla_{\boldsymbol{\lambda}} \mathcal{D}_\rho = \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x}), \qquad \boldsymbol{q}(\boldsymbol{\lambda}; \boldsymbol{x}) \in \partial h(\boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x})).$$

  The $\boldsymbol{\lambda}$- and $\boldsymbol{\mu}$-updates are convex combinations, hence the actual dual orbit is bounded without algorithmic projection.

▸ Lemmas C1–C2

- **Module 2: frozen three-value descent.** Freeze $(r_k, \boldsymbol{\mu}^k)$ and identify the $(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{z})$ steps with Perturbed Smoothed GDA. Its three-value Lyapunov satisfies

$$\Delta \Psi_k \leq -c_x \|\Delta \boldsymbol{x}^{k+1}\|^2 - c_\lambda e_{\lambda,k}^2 - c_z \beta_k \|\boldsymbol{z}^k - \boldsymbol{x}^{k+1}\|^2.$$

▸ Lemmas C3–C4

Core mechanism

The three-value Lyapunov controls an ideal dual step directly and avoids the $s_k^3 \|\Delta \boldsymbol{x}^{k+1}\|^2$ loss caused by exact-target tracking.

Proof Sketch II: Moving Parameters and Rate Extraction

- **Module 3: summable transport and master budget.** Moving $\boldsymbol{\mu}^k$ and varying r_k contribute only $O(r_k\gamma_k)$ and $O(|r_{k+1}-r_k|)$. Both sequences are summable, so frozen descent telescopes to

$$\sum_{k=0}^{\infty}\left[\|\Delta \boldsymbol{x}^{k+1}\|^2 + e_{\lambda,k}^2 + \beta_k\|\boldsymbol{z}^k - \boldsymbol{x}^{k+1}\|^2\right] < \infty.$$

▸ Lemmas C5–C6

- **Module 4: split-KKT reconstruction and tail balance.** At one common $\kappa \in \{\lceil K/2\rceil, \dots, K-1\}$,

$$\|\Delta \boldsymbol{x}^{\kappa+1}\| + e_{\lambda,\kappa} = O(K^{-1/2}), \qquad \|\boldsymbol{z}^{\kappa} - \boldsymbol{x}^{\kappa+1}\| + r_\kappa\|\boldsymbol{\lambda}^{\kappa} - \boldsymbol{\mu}^{\kappa}\| = O(K^{-1/3}).$$

The update identities convert these quantities into stationarity and feasibility residuals of order $O(K^{-1/3})$.

▸ Lemmas C7–C8

Final rate

The exact subgradient residual is zero; the other two split-KKT residuals are $O(K^{-1/3})$. Therefore $K = O(\varepsilon^{-3})$.

Candidate Proof Status and Scope

- The four-block algorithm on the slides is exactly the algorithm used in the varying-s_k proof.
- The frozen-center step is the translated fixed-perturbation instance of Li–Nagarajan–Pan–Zhang's Perturbed Smoothed GDA three-value analysis [4].
- The moving-$\boldsymbol{\mu}$ transport, varying-r_k drift, no-projection boundedness, and split-KKT reconstruction are proved separately in the complete-proof appendix.
- The theorem requires globally Lipschitz h and gives a deterministic tail-best-iterate split-KKT result.
- The result does not establish the stronger constant-s route and does not claim last-iterate complexity.

VI. References

References

[1] H. Zhang, N. Marumo, T. K. Pong, and A. Takeda, "Complexity and Convergence Analysis of a Single-Loop SDCAM for Lipschitz Composite Optimization and Beyond," arXiv:2512.24059, 2025.

[2] K. Sun and X. A. Sun, "Dual Descent Augmented Lagrangian Method and Alternating Direction Method of Multipliers," *SIAM Journal on Optimization*, 34(2):1679–1707, 2024.

[3] J. M. Danskin, "The Theory of Max–Min, with Applications," *SIAM Journal on Applied Mathematics*, 14(4):641–664, 1966.

[4] J. Li, M. Nagarajan, S. Pan, and N. Zhang, "Smoothing Meets Perturbation: Unified and Tight Analysis for Nonconvex–Concave Minimax Optimization," arXiv:2602.14185v2, 2026.

Appendix: Complete Proofs

1. Lemmas C1–C2: reduced geometry and bounded dual orbit
2. Lemmas C3–C4: three-value Lyapunov and frozen descent
3. Lemmas C5–C6: moving-parameter transport and master budget
4. Lemmas C7–C8: residual reconstruction and tail extraction
5. Main Theorem: complete complexity proof

Detail level

Every appendix result includes its assumptions, full derivation, and the exact point where it enters the main theorem.

Appendix: Lemma C1.1—Reduced-Payoff Calculus

Contents

Define

$$\boldsymbol{y}(\boldsymbol{\lambda};\boldsymbol{x}) := \operatorname{prox}_{h/\rho}\left(\mathcal{A}(\boldsymbol{x}) + \frac{\boldsymbol{\lambda}}{\rho}\right), \qquad \boldsymbol{q}(\boldsymbol{\lambda};\boldsymbol{x}) := \boldsymbol{\lambda} + \rho[\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{\lambda};\boldsymbol{x})],$$

and

$$\mathcal{D}_\rho(\boldsymbol{x},\boldsymbol{\lambda}) := \min_{\boldsymbol{y}}\left\{f(\boldsymbol{x}) + h(\boldsymbol{y}) + \langle\boldsymbol{\lambda}, \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\rangle + \frac{\rho}{2}\|\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\|^2\right\}.$$

Then

$$\nabla_{\boldsymbol{x}}\mathcal{D}_\rho = \nabla f(\boldsymbol{x}) + J_{\mathcal{A}}(\boldsymbol{x})^\top \boldsymbol{q}(\boldsymbol{\lambda};\boldsymbol{x}), \qquad \nabla_{\boldsymbol{\lambda}}\mathcal{D}_\rho = \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{\lambda};\boldsymbol{x}),$$

and

$$\boldsymbol{q}(\boldsymbol{\lambda};\boldsymbol{x}) \in \partial h(\boldsymbol{y}(\boldsymbol{\lambda};\boldsymbol{x})), \qquad \|\boldsymbol{q}(\boldsymbol{\lambda};\boldsymbol{x})\| \le L_h.$$

Proof. Completing the square gives

$$h(\boldsymbol{y}) + \frac{\rho}{2}\left\|\boldsymbol{y} - \mathcal{A}(\boldsymbol{x}) - \frac{\boldsymbol{\lambda}}{\rho}\right\|^2 - \frac{1}{2\rho}\|\boldsymbol{\lambda}\|^2.$$

Thus the minimizer is unique. Its optimality condition is

$$0 \in \partial h(\boldsymbol{y}) + \rho[\boldsymbol{y} - \mathcal{A}(\boldsymbol{x})] - \boldsymbol{\lambda},$$

which gives the exact certificate. Global Lipschitz continuity of h gives $\|\boldsymbol{q}\| \le L_h$. Danskin's theorem applied to the unique minimizer gives the two displayed gradients.

Appendix: Lemma C1.2—Joint Smoothness

Contents

Moreau decomposition yields

$$\boldsymbol{q}(\boldsymbol{\lambda};\boldsymbol{x}) = \operatorname{prox}_{\rho h^*}(\boldsymbol{\lambda} + \rho\mathcal{A}(\boldsymbol{x})).$$

Hence nonexpansiveness gives

$$\|\boldsymbol{q}(\boldsymbol{\lambda};\boldsymbol{x}) - \boldsymbol{q}(\boldsymbol{\lambda}';\boldsymbol{x}')\| \le \|\boldsymbol{\lambda} - \boldsymbol{\lambda}'\| + \rho G_{\mathcal{A}}\|\boldsymbol{x} - \boldsymbol{x}'\|.$$

Using $\|\boldsymbol{q}\| \le L_h$, $\|J_{\mathcal{A}}\| \le G_{\mathcal{A}}$, and the Lipschitz constants of ∇f and $J_{\mathcal{A}}$,

$$\begin{aligned}&\|\nabla_{\boldsymbol{x}}\mathcal{D}_\rho(\boldsymbol{x},\boldsymbol{\lambda}) - \nabla_{\boldsymbol{x}}\mathcal{D}_\rho(\boldsymbol{x}',\boldsymbol{\lambda}')\| \\ &\quad\le (L_f + L_{\mathcal{A}}L_h + \rho G_{\mathcal{A}}^2)\|\boldsymbol{x} - \boldsymbol{x}'\| + G_{\mathcal{A}}\|\boldsymbol{\lambda} - \boldsymbol{\lambda}'\|.\end{aligned}$$

Moreover,

$$\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{\lambda};\boldsymbol{x}) = \rho^{-1}[\boldsymbol{q}(\boldsymbol{\lambda};\boldsymbol{x}) - \boldsymbol{\lambda}].$$

Firm nonexpansiveness makes $I - \operatorname{prox}_{\rho h^*}$ nonexpansive. Therefore

$$\|\nabla_{\boldsymbol{\lambda}}\mathcal{D}_\rho(\boldsymbol{x},\boldsymbol{\lambda}) - \nabla_{\boldsymbol{\lambda}}\mathcal{D}_\rho(\boldsymbol{x}',\boldsymbol{\lambda}')\| \le G_{\mathcal{A}}\|\boldsymbol{x} - \boldsymbol{x}'\| + \rho^{-1}\|\boldsymbol{\lambda} - \boldsymbol{\lambda}'\|.$$

Thus $\mathcal{D}_\rho$ is jointly ℓ_ρ-smooth for $\ell_\rho := L_f + L_{\mathcal{A}}L_h + \rho G_{\mathcal{A}}^2 + 2G_{\mathcal{A}} + \rho^{-1}$.

Appendix: Lemma C1.3—Concavity and Exact Maximum

Contents

For fixed $\boldsymbol{x}$, $\mathcal{D}_\rho(\boldsymbol{x}, \cdot)$ is the pointwise infimum of affine functions of $\boldsymbol{\lambda}$, hence it is concave. It also satisfies

$$\max_{\boldsymbol{\lambda} \in \mathbb{R}^m} \mathcal{D}_\rho(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) + h(\mathcal{A}(\boldsymbol{x})).$$

Proof. Taking $\boldsymbol{y} = \mathcal{A}(\boldsymbol{x})$ in the inner minimization gives, for every $\boldsymbol{\lambda}$,

$$\mathcal{D}_\rho(\boldsymbol{x}, \boldsymbol{\lambda}) \leq f(\boldsymbol{x}) + h(\mathcal{A}(\boldsymbol{x})).$$

Because h is finite-valued and convex, choose $\bar{\boldsymbol{\lambda}} \in \partial h(\mathcal{A}(\boldsymbol{x}))$. Then $\boldsymbol{y} = \mathcal{A}(\boldsymbol{x})$ satisfies

$$0 \in \partial h(\boldsymbol{y}) - \bar{\boldsymbol{\lambda}} + \rho[\boldsymbol{y} - \mathcal{A}(\boldsymbol{x})],$$

so it is the inner minimizer and equality holds at $\bar{\boldsymbol{\lambda}}$. Global Lipschitz continuity gives $\|\bar{\boldsymbol{\lambda}}\| \leq L_h$. Consequently, every maximization needed later may select a maximizer in the L_h-ball.

Use in the proof

This exact maximum identity supplies the uniform lower bound for the three-value Lyapunov function in Lemma C6.

Appendix: Lemma C2—No-Projection Boundedness

Contents

Let $M := \max\{\|\boldsymbol{\lambda}^0\|, \|\boldsymbol{\mu}^0\|, L_h\}$. If $\alpha \le (\rho^{-1} + r_0)^{-1}$ and $0 \le \gamma_k \le 1$, then

$$\|\boldsymbol{\lambda}^k\| \le M, \qquad \|\boldsymbol{\mu}^k\| \le M, \qquad k \ge 0.$$

Proof. Set

$$\widehat{\boldsymbol{q}}^{k+1} := \boldsymbol{\lambda}^k + \rho[\mathcal{A}(\boldsymbol{x}^{k+1}) - \widehat{\boldsymbol{y}}^{k+1}].$$

Lemma C1 gives $\|\widehat{\boldsymbol{q}}^{k+1}\| \le L_h$. Substitution into the $\boldsymbol{\lambda}$-update yields

$$\boldsymbol{\lambda}^{k+1} = [1 - \alpha(\rho^{-1} + r_k)]\boldsymbol{\lambda}^k + \frac{\alpha}{\rho}\widehat{\boldsymbol{q}}^{k+1} + \alpha r_k \boldsymbol{\mu}^k.$$

Since $0 < r_k \le r_0$, the three coefficients are nonnegative and sum to one. Thus, whenever $\|\boldsymbol{\lambda}^k\|, \|\boldsymbol{\mu}^k\| \le M$,

$$\|\boldsymbol{\lambda}^{k+1}\| \le M.$$

The update $\boldsymbol{\mu}^{k+1} = (1 - \gamma_k)\boldsymbol{\mu}^k + \gamma_k \boldsymbol{\lambda}^{k+1}$ then gives $\|\boldsymbol{\mu}^{k+1}\| \le M$. The base case follows from the definition of M, so induction proves the claim. Define $B_M := \{v : \|v\| \le M\}$. Choose $R > M$ and $Y := \{\boldsymbol{\lambda} : \|\boldsymbol{\lambda}\| \le R\}$. The actual iterates lie strictly inside Y; projection onto Y is used only in the proof.

Appendix: Lemma C3.1—Three Value Functions

Contents

For fixed $r \in (0, r_0]$, $\boldsymbol{\mu} \in B_M$, and $\boldsymbol{z} \in \mathbb{R}^n$, define

$$\mathcal{F}_{r,\boldsymbol{\mu}}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{z}) := \mathcal{D}_\rho(\boldsymbol{x}, \boldsymbol{\lambda}) + \frac{r_x}{2}\|\boldsymbol{x} - \boldsymbol{z}\|^2 - \frac{r}{2}\|\boldsymbol{\lambda} - \boldsymbol{\mu}\|^2,$$

$$d_{r,\boldsymbol{\mu}}(\boldsymbol{\lambda}, \boldsymbol{z}) := \min_{\boldsymbol{x}} \mathcal{F}_{r,\boldsymbol{\mu}}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{z}), \qquad p_{r,\boldsymbol{\mu}}(\boldsymbol{z}) := \min_{\boldsymbol{x}} \max_{\boldsymbol{\lambda} \in Y} \mathcal{F}_{r,\boldsymbol{\mu}}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{z}).$$

The minima exist, and the minimizer defining $d_{r,\boldsymbol{\mu}}$ is unique. **Proof.** Joint ℓ_ρ-smoothness implies that $\mathcal{D}_\rho(\cdot, \boldsymbol{\lambda})$ is ℓ_ρ-weakly convex. Therefore

$$\mathcal{F}_{r,\boldsymbol{\mu}}(\cdot, \boldsymbol{\lambda}, \boldsymbol{z}) \text{ is } (r_x - \ell_\rho)\text{-strongly convex.}$$

It is finite and coercive, so its unique minimizer exists. For each $\boldsymbol{\lambda} \in Y$, the same strong-convexity modulus holds. The pointwise maximum over compact Y preserves strong convexity, finiteness, and coercivity. Hence the minimum defining $p_{r,\boldsymbol{\mu}}$ also exists.

Appendix: Lemma C3.2—Three-Value Lyapunov

Contents

Define

$$\Psi_{r,\boldsymbol{\mu}}(\boldsymbol{x},\boldsymbol{\lambda},\boldsymbol{z}) := \mathcal{F}_{r,\boldsymbol{\mu}}(\boldsymbol{x},\boldsymbol{\lambda},\boldsymbol{z}) - 2d_{r,\boldsymbol{\mu}}(\boldsymbol{\lambda},\boldsymbol{z}) + 2p_{r,\boldsymbol{\mu}}(\boldsymbol{z}).$$

Its exact decomposition is

$$\Psi_{r,\boldsymbol{\mu}} = (\mathcal{F}_{r,\boldsymbol{\mu}} - d_{r,\boldsymbol{\mu}}) + (p_{r,\boldsymbol{\mu}} - d_{r,\boldsymbol{\mu}}) + p_{r,\boldsymbol{\mu}}.$$

The first two bracketed terms are nonnegative because d is a pointwise minimum and

$$p_{r,\boldsymbol{\mu}}(\boldsymbol{z}) = \min_{\boldsymbol{x}} \max_{\boldsymbol{\lambda}' \in Y} \mathcal{F}_{r,\boldsymbol{\mu}}(\boldsymbol{x},\boldsymbol{\lambda}',\boldsymbol{z}) \geq d_{r,\boldsymbol{\mu}}(\boldsymbol{\lambda},\boldsymbol{z}).$$

Let

$$\boldsymbol{x}_{r,\boldsymbol{\mu}}(\boldsymbol{\lambda},\boldsymbol{z}) := \arg\min_{\boldsymbol{x}} \mathcal{F}_{r,\boldsymbol{\mu}}(\boldsymbol{x},\boldsymbol{\lambda},\boldsymbol{z})$$

and define the ideal dual step

$$\boldsymbol{\lambda}_+(\boldsymbol{z}) := \mathrm{proj}_Y \left[\boldsymbol{\lambda} + \alpha \nabla_{\boldsymbol{\lambda}} \mathcal{F}_{r,\boldsymbol{\mu}}(\boldsymbol{x}_{r,\boldsymbol{\mu}}(\boldsymbol{\lambda},\boldsymbol{z}),\boldsymbol{\lambda},\boldsymbol{z})\right].$$

The three terms of Ψ respectively measure primal suboptimality, dual ascent error, and the proximal-center value.

Appendix: Lemma C4.1—Frozen-Center Transfer

Contents

Fix $(r, \boldsymbol{\mu})$ and set

$$\boldsymbol{\nu} := \boldsymbol{\lambda} - \boldsymbol{\mu}, \qquad \widetilde{D}_{\boldsymbol{\mu}}(\boldsymbol{x}, \boldsymbol{\nu}) := \mathcal{D}_\rho(\boldsymbol{x}, \boldsymbol{\nu} + \boldsymbol{\mu}).$$

Then $\widetilde{D}_{\boldsymbol{\mu}}$ is ℓ_ρ-smooth and concave in $\boldsymbol{\nu}$. The translated dual set $Y - \boldsymbol{\mu}$ is compact and convex, contains the origin because $\boldsymbol{\mu} \in B_M \subset Y$, and has the same diameter as Y. Moreover,

$$\mathcal{F}_{r,\boldsymbol{\mu}} = \widetilde{D}_{\boldsymbol{\mu}}(\boldsymbol{x}, \boldsymbol{\nu}) + \frac{r_x}{2}\|\boldsymbol{x} - \boldsymbol{z}\|^2 - \frac{r}{2}\|\boldsymbol{\nu}\|^2.$$

Thus the frozen $(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{z})$ update is exactly Perturbed Smoothed GDA on the translated problem. Li–Nagarajan–Pan–Zhang [4], Lemmas 4.4–4.5, give

$$\begin{aligned}\Psi^+ - \Psi \le & -\frac{1}{8c}\|\boldsymbol{x}^+ - \boldsymbol{x}\|^2 - \frac{1}{8\alpha}\|\boldsymbol{\lambda} - \boldsymbol{\lambda}_+(\boldsymbol{z})\|^2 - \frac{r_x\beta}{8}\|\boldsymbol{z} - \boldsymbol{x}^+\|^2 \\ & + 24 r_x \beta \|\boldsymbol{x}_r^*(\boldsymbol{z}) - \boldsymbol{x}_r(\boldsymbol{\lambda}_+(\boldsymbol{z}), \boldsymbol{z})\|^2,\end{aligned}$$

and

$$\|\boldsymbol{x}_r^*(\boldsymbol{z}) - \boldsymbol{x}_r(\boldsymbol{\lambda}_+(\boldsymbol{z}), \boldsymbol{z})\| \le \Omega(r)\|\boldsymbol{\lambda} - \boldsymbol{\lambda}_+(\boldsymbol{z})\|.$$

The perturbation-change term in their Lemma 4.4 is zero because r is frozen.

Appendix: Lemma C4.2—Parameter Map and Absorption

Contents

The parameter map to Condition 3.2 of [4] is

$$(\ell, r_x, c, \alpha, \beta, r_y) \longmapsto (\ell_\rho, r_x, c, \alpha, \beta, r).$$

Its dual error coefficient becomes

$$\Omega(r) := \frac{1}{\sqrt{r}} \frac{(r_x - \ell_\rho - r) + \alpha(\ell_\rho + r)(3r_x - 2\ell_\rho - 2r)}{\alpha(r_x - \ell_\rho - r)^{3/2}}.$$

Assume $r_x > 3\ell_\rho$, $0 < r \le r_0 \le \ell_\rho$, the stated bounds on c and α, and

$$0 < \beta \le \min\left\{ \frac{1}{36}, \frac{(r_x - \ell_\rho - r)^2}{384 r_x (r_x + \ell_\rho + r)^2}, \frac{1}{384 r_x \alpha \Omega(r)^2} \right\}.$$

Then

$$24 r_x \beta \Omega(r)^2 \le \frac{1}{16\alpha}.$$

Substitution in Lemma C4.1 gives the complete frozen descent inequality

$$\Psi^+ - \Psi \le -\frac{1}{8c}\|\boldsymbol{x}^+ - \boldsymbol{x}\|^2 - \frac{1}{16\alpha}\|\boldsymbol{\lambda} - \boldsymbol{\lambda}_+(\boldsymbol{z})\|^2 - \frac{r_x \beta}{8}\|\boldsymbol{z} - \boldsymbol{x}^+\|^2.$$

Since $\Omega(r) \le C_\Omega/\sqrt{r}$, choosing $\beta_k = (\theta_z/\ell_\rho) r_k$ with sufficiently small fixed θ_z makes these bounds uniform in k.

Appendix: Lemma C5.1—Moving-Center Transport

Contents

For $\|\boldsymbol{\mu}\|, \|\boldsymbol{\mu}'\| \le M$, $\boldsymbol{\lambda} \in Y$, and $u = (\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{z})$,

$$|\Psi_{r,\boldsymbol{\mu}'}(u) - \Psi_{r,\boldsymbol{\mu}}(u)| \le 5r(R+M)\|\boldsymbol{\mu}' - \boldsymbol{\mu}\| + \frac{5r}{2}\|\boldsymbol{\mu}' - \boldsymbol{\mu}\|^2.$$

Proof. Let $\delta\boldsymbol{\mu} := \boldsymbol{\mu}' - \boldsymbol{\mu}$. The only center-dependent term satisfies

$$-\frac{r}{2}\|\boldsymbol{\lambda} - \boldsymbol{\mu}'\|^2 + \frac{r}{2}\|\boldsymbol{\lambda} - \boldsymbol{\mu}\|^2 = r\langle \boldsymbol{\lambda} - \boldsymbol{\mu}, \delta\boldsymbol{\mu}\rangle - \frac{r}{2}\|\delta\boldsymbol{\mu}\|^2.$$

Because $\|\boldsymbol{\lambda}\| \le R$ and $\|\boldsymbol{\mu}\| \le M$,

$$\sup_{\boldsymbol{x},\boldsymbol{\lambda},\boldsymbol{z}} |\mathcal{F}_{r,\boldsymbol{\mu}'} - \mathcal{F}_{r,\boldsymbol{\mu}}| \le r(R+M)\|\delta\boldsymbol{\mu}\| + \frac{r}{2}\|\delta\boldsymbol{\mu}\|^2 =: \delta.$$

Uniform perturbation by δ changes each minimum, maximum, and min–max value by at most δ. Since $\Psi = \mathcal{F} - 2d + 2p$, the total change is at most 5δ. For the algorithm,

$$\|\boldsymbol{\mu}^{k+1} - \boldsymbol{\mu}^k\| = \gamma_k \|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^k\| \le 2M\gamma_k.$$

Using $\gamma_k \le 1$, the transport is bounded by $C_\mu r_k \gamma_k$.

Appendix: Lemma C5.2—Perturbation-Scale Drift

Contents

For fixed $\boldsymbol{\mu}$ and any $r, r' > 0$,

$$|\Psi_{r',\boldsymbol{\mu}}(u) - \Psi_{r,\boldsymbol{\mu}}(u)| \le C_r |r' - r|, \qquad C_r := \frac{5}{2}(R+M)^2.$$

Proof. The surrogate difference is exactly

$$\mathcal{F}_{r',\boldsymbol{\mu}}(u) - \mathcal{F}_{r,\boldsymbol{\mu}}(u) = -\frac{r'-r}{2}\|\boldsymbol{\lambda} - \boldsymbol{\mu}\|^2.$$

Therefore its uniform magnitude is at most $\frac{1}{2}(R+M)^2|r'-r|$. Applying the same value-perturbation argument as in Lemma C5.1 and the coefficient sum $1+2+2=5$ proves the claim.
For

$$r_k = c_r(k+k_0)^{-1/3}, \qquad \gamma_k = \frac{\theta_\mu}{k+k_0},$$

r_k is decreasing to zero, hence

$$\sum_{k=0}^{\infty} |r_{k+1} - r_k| = r_0.$$

Also

$$r_k\gamma_k = c_r\theta_\mu(k+k_0)^{-4/3}, \qquad \sum_{k=0}^{\infty} r_k\gamma_k < \infty.$$

Thus both moving-parameter errors are summable with constants independent of K and ε.

Appendix: Lemma C6.1—Master One-Step Inequality

Contents

Let

$$u^k := (\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{z}^k), \quad V_k := \Psi_{r_k, \boldsymbol{\mu}^k}(u^k), \quad e_{\lambda,k} := \|\boldsymbol{\lambda}^k - \boldsymbol{\lambda}_{+,k}(\boldsymbol{z}^k)\|.$$

Insert two intermediate values:

$$\begin{aligned} V_{k+1} - V_k = &[\Psi_{r_k, \boldsymbol{\mu}^k}(u^{k+1}) - \Psi_{r_k, \boldsymbol{\mu}^k}(u^k)] \\ &+ [\Psi_{r_k, \boldsymbol{\mu}^{k+1}}(u^{k+1}) - \Psi_{r_k, \boldsymbol{\mu}^k}(u^{k+1})] \\ &+ [\Psi_{r_{k+1}, \boldsymbol{\mu}^{k+1}}(u^{k+1}) - \Psi_{r_k, \boldsymbol{\mu}^{k+1}}(u^{k+1})]. \end{aligned}$$

Lemma C4 controls the first bracket. Lemma C5.1 controls the second, and Lemma C5.2 controls the third. Hence

$$\begin{aligned} V_{k+1} - V_k \le &-\frac{1}{8c}\|\Delta \boldsymbol{x}^{k+1}\|^2 - \frac{1}{16\alpha} e_{\lambda,k}^2 - \frac{r_x \beta_k}{8}\|\boldsymbol{z}^k - \boldsymbol{x}^{k+1}\|^2 \\ &+ C_r |r_{k+1} - r_k| + C_\mu r_k \gamma_k. \end{aligned}$$

Every positive error on the right is summable by Lemma C5.2.

Appendix: Lemma C6.2—Uniform Lower Bound

Contents

For every k,

$$V_k \geq \underline{V} := F_{\inf} - \frac{r_0}{2}(L_h + M)^2.$$

Proof. Lemma C3 gives

$$\Psi_{r,\boldsymbol{\mu}} = (\mathcal{F} - d) + (p - d) + p \geq p.$$

For each $\boldsymbol{x}$, Lemma C1.3 provides $\bar{\boldsymbol{\lambda}}(\boldsymbol{x}) \in \partial h(\mathcal{A}(\boldsymbol{x})) \subseteq Y$ such that

$$\mathcal{D}_\rho(\boldsymbol{x}, \bar{\boldsymbol{\lambda}}(\boldsymbol{x})) = f(\boldsymbol{x}) + h(\mathcal{A}(\boldsymbol{x})), \qquad \|\bar{\boldsymbol{\lambda}}(\boldsymbol{x})\| \leq L_h.$$

Therefore

$$\begin{aligned}\max_{\boldsymbol{\lambda} \in Y} \mathcal{F}_{r,\boldsymbol{\mu}}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{z}) &\geq f(\boldsymbol{x}) + h(\mathcal{A}(\boldsymbol{x})) + \frac{r_x}{2}\|\boldsymbol{x} - \boldsymbol{z}\|^2 - \frac{r}{2}\|\bar{\boldsymbol{\lambda}}(\boldsymbol{x}) - \boldsymbol{\mu}\|^2 \\ &\geq F_{\inf} - \frac{r_0}{2}(L_h + M)^2.\end{aligned}$$

Taking the minimum over $\boldsymbol{x}$ proves the same lower bound for p, and hence for V_k.

Appendix: Lemma C6.3—Finite Master Budget

Contents

Sum Lemma C6.1 from $k = 0$ to $N - 1$. After telescoping and applying Lemma C6.2,

$$\sum_{k=0}^{N-1} \left[\frac{1}{8c} \|\Delta \boldsymbol{x}^{k+1}\|^2 + \frac{1}{16\alpha} e_{\lambda,k}^2 + \frac{r_x \beta_k}{8} \|\boldsymbol{z}^k - \boldsymbol{x}^{k+1}\|^2 \right]$$

$$\leq V_0 - \underline{V} + C_r \sum_{k=0}^{\infty} |r_{k+1} - r_k| + C_\mu \sum_{k=0}^{\infty} r_k \gamma_k .$$

The right-hand side is a finite constant independent of N, K, and ε. Letting $N \to \infty$ and absorbing the fixed positive coefficients gives a constant $B < \infty$ such that

$$\sum_{k=0}^{\infty} \left[\|\Delta \boldsymbol{x}^{k+1}\|^2 + e_{\lambda,k}^2 + \beta_k \|\boldsymbol{z}^k - \boldsymbol{x}^{k+1}\|^2 \right] \leq B.$$

Use in the proof

This horizon-independent budget is the sole averaging input in the tail-index argument. No last-iterate claim is used.

Appendix: Lemma C7.1—Primal Target Error

Contents

Let

$$\boldsymbol{x}_k^* := \arg\min_{\boldsymbol{x}} \mathcal{F}_{r_k,\boldsymbol{\mu}^k}(\boldsymbol{x}, \boldsymbol{\lambda}^k, \boldsymbol{z}^k), \qquad m_x := r_x - \ell_\rho > 0.$$

Then

$$\|\boldsymbol{x}^{k+1} - \boldsymbol{x}_k^*\| \le \left(1 + \frac{1}{cm_x}\right) \|\Delta \boldsymbol{x}^{k+1}\|.$$

Proof. The $\boldsymbol{x}$-subproblem is m_x-strongly convex and $\nabla_{\boldsymbol{x}} \mathcal{F}_{r_k,\boldsymbol{\mu}^k}(\boldsymbol{x}_k^*, \boldsymbol{\lambda}^k, \boldsymbol{z}^k) = 0$. Strong monotonicity gives

$$m_x \|\boldsymbol{x}^k - \boldsymbol{x}_k^*\| \le \|\nabla_{\boldsymbol{x}} \mathcal{F}_{r_k,\boldsymbol{\mu}^k}(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{z}^k)\|.$$

The algorithmic $\boldsymbol{x}$-step yields

$$\nabla_{\boldsymbol{x}} \mathcal{F}_{r_k,\boldsymbol{\mu}^k}(\boldsymbol{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{z}^k) = -c^{-1} \Delta \boldsymbol{x}^{k+1}.$$

Thus $\|\boldsymbol{x}^k - \boldsymbol{x}_k^*\| \le (cm_x)^{-1} \|\Delta \boldsymbol{x}^{k+1}\|$. The triangle inequality with $\boldsymbol{x}^{k+1} - \boldsymbol{x}_k^* = (\boldsymbol{x}^{k+1} - \boldsymbol{x}^k) + (\boldsymbol{x}^k - \boldsymbol{x}_k^*)$ proves the claim.

Appendix: Lemma C7.2—Actual Dual Increment

Contents

Because Lemma C2 keeps the raw dual update in $B_M \subset Y$,

$$\boldsymbol{\lambda}^{k+1} = \mathrm{proj}_Y \left[\boldsymbol{\lambda}^k + \alpha \nabla_{\boldsymbol{\lambda}} \mathcal{F}_{r_k, \boldsymbol{\mu}^k} (\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k, \boldsymbol{z}^k) \right].$$

The ideal step is the same expression with $\boldsymbol{x}^{k+1}$ replaced by $\boldsymbol{x}_k^*$. Projection nonexpansiveness and the cross-Lipschitz bound in Lemma C1 give

$$\begin{aligned} \|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\lambda}_{+,k}(\boldsymbol{z}^k)\| &\le \alpha G_{\mathcal{A}} \|\boldsymbol{x}^{k+1} - \boldsymbol{x}_k^*\| \\ &\le \alpha G_{\mathcal{A}} \left(1 + \frac{1}{c m_x}\right) \|\Delta \boldsymbol{x}^{k+1}\|. \end{aligned}$$

Adding and subtracting $\boldsymbol{\lambda}_{+,k}(\boldsymbol{z}^k)$ yields

$$\|\boldsymbol{\lambda}^{k+1} - \boldsymbol{\lambda}^k\| \le e_{\lambda,k} + C_{\lambda x} \|\Delta \boldsymbol{x}^{k+1}\|,$$

where

$$C_{\lambda x} := \alpha G_{\mathcal{A}} \left(1 + \frac{1}{c m_x}\right).$$

This proves that the ideal dual error controlled by the Lyapunov budget also controls the actual multiplier increment.

Appendix: Lemma C8.1—Exact Certificate and Stationarity

Contents

Define

$$\boldsymbol{y}_k^{\mathrm{out}} := \widehat{\boldsymbol{y}}^{k+1}, \qquad \boldsymbol{q}_k^{\mathrm{out}} := \boldsymbol{\lambda}^k + \rho[\mathcal{A}(\boldsymbol{x}^{k+1}) - \boldsymbol{y}_k^{\mathrm{out}}].$$

Lemma C1 gives the exact relation

$$\boldsymbol{q}_k^{\mathrm{out}} \in \partial h(\boldsymbol{y}_k^{\mathrm{out}}).$$

The $\boldsymbol{x}$-step is equivalent to

$$\nabla_{\boldsymbol{x}} \mathcal{D}_\rho(\boldsymbol{x}^k, \boldsymbol{\lambda}^k) + r_x(\boldsymbol{x}^k - \boldsymbol{z}^k) = -c^{-1}\Delta\boldsymbol{x}^{k+1}.$$

Since

$$\nabla_{\boldsymbol{x}} \mathcal{D}_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^k) = \nabla f(\boldsymbol{x}^{k+1}) + J_{\mathcal{A}}(\boldsymbol{x}^{k+1})^\top \boldsymbol{q}_k^{\mathrm{out}},$$

adding and subtracting the gradient at $\boldsymbol{x}^k$, applying the fixed-$\boldsymbol{\lambda}$ a_ρ-smoothness bound, and using $\boldsymbol{x}^k - \boldsymbol{z}^k = (\boldsymbol{x}^{k+1} - \boldsymbol{z}^k) - \Delta\boldsymbol{x}^{k+1}$ give

$$\begin{aligned}
&\|\nabla f(\boldsymbol{x}^{k+1}) + J_{\mathcal{A}}(\boldsymbol{x}^{k+1})^\top \boldsymbol{q}_k^{\mathrm{out}}\| \\
&\quad \le (c^{-1} + r_x + a_\rho)\|\Delta\boldsymbol{x}^{k+1}\| + r_x\|\boldsymbol{x}^{k+1} - \boldsymbol{z}^k\|.
\end{aligned}$$

Appendix: Lemma C8.2—Feasibility Reconstruction

Contents

The dual update gives the exact identity

$$\frac{\boldsymbol{\lambda}^{k+1}-\boldsymbol{\lambda}^{k}}{\alpha}=\mathcal{A}(\boldsymbol{x}^{k+1})-\boldsymbol{y}_{k}^{\mathrm{out}}-r_{k}(\boldsymbol{\lambda}^{k}-\boldsymbol{\mu}^{k}).$$

Rearranging and using Lemma C2,

$$\begin{aligned}\|\mathcal{A}(\boldsymbol{x}^{k+1})-\boldsymbol{y}_{k}^{\mathrm{out}}\| &\le \alpha^{-1}\|\boldsymbol{\lambda}^{k+1}-\boldsymbol{\lambda}^{k}\|+r_{k}\|\boldsymbol{\lambda}^{k}-\boldsymbol{\mu}^{k}\| \\ &\le \alpha^{-1}\|\boldsymbol{\lambda}^{k+1}-\boldsymbol{\lambda}^{k}\|+2Mr_{k}.\end{aligned}$$

Combining this inequality with Lemma C7.2 gives

$$\|\mathcal{A}(\boldsymbol{x}^{k+1})-\boldsymbol{y}_{k}^{\mathrm{out}}\|\le\alpha^{-1}e_{\lambda,k}+\alpha^{-1}C_{\lambda x}\|\Delta\boldsymbol{x}^{k+1}\|+2Mr_{k}.$$

Rate mechanism

The finite-time feasibility bias is controlled by $r_k \downarrow 0$ and the bounded center gap. The proof does not require a rate for $\|\boldsymbol{\lambda}^{k}-\boldsymbol{\mu}^{k}\|$.

Appendix: Lemma C8.3—Common Tail Index

Contents

Let $I_K := \{\lceil K/2 \rceil, \ldots, K-1\}$. For $K \geq 4$, $|I_K| \geq K/4$. Lemma C6.3 therefore provides one common $\kappa \in I_K$ such that

$$\|\Delta \boldsymbol{x}^{\kappa+1}\|^2 + e_{\lambda,\kappa}^2 + \beta_\kappa \|\boldsymbol{z}^\kappa - \boldsymbol{x}^{\kappa+1}\|^2 \leq \frac{4B}{K}.$$

Consequently,

$$\|\Delta \boldsymbol{x}^{\kappa+1}\| + e_{\lambda,\kappa} = O(K^{-1/2}).$$

Because $\beta_\kappa = \Theta(K^{-1/3})$ on the tail,

$$\|\boldsymbol{z}^\kappa - \boldsymbol{x}^{\kappa+1}\| \leq \sqrt{\frac{4B}{K\beta_\kappa}} = O(K^{-1/3}).$$

Lemma C7.2 gives

$$\|\boldsymbol{\lambda}^{\kappa+1} - \boldsymbol{\lambda}^\kappa\| = O(K^{-1/2}).$$

Finally, $r_\kappa = \Theta(K^{-1/3})$. Lemmas C8.1–C8.2 now imply that stationarity and feasibility are both $O(K^{-1/3})$ at this same index.

Appendix: Main Theorem—Statement

Contents

Assume the hypotheses on f, $\mathcal{A}$, and h, and choose the parameters as stated in the Candidate algorithm. Then for every $K \geq 4$, there exists

$$\kappa \in \{\lceil K/2 \rceil, \ldots, K-1\}$$

such that

$$\boldsymbol{q}_\kappa^{\mathrm{out}} \in \partial h(\boldsymbol{y}_\kappa^{\mathrm{out}}),$$
$$\|\nabla f(\boldsymbol{x}^{\kappa+1}) + J_{\mathcal{A}}(\boldsymbol{x}^{\kappa+1})^\top \boldsymbol{q}_\kappa^{\mathrm{out}}\| \leq C_{\mathrm{stat}} K^{-1/3},$$

and

$$\|\mathcal{A}(\boldsymbol{x}^{\kappa+1}) - \boldsymbol{y}_\kappa^{\mathrm{out}}\| \leq C_{\mathrm{feas}} K^{-1/3}.$$

The constants are independent of K and the requested tolerance ε.

Appendix: Main Theorem—Complete Proof

Contents

Proof. Lemmas C1–C2 establish the exact output certificate and bounded dual orbit. Lemmas C3–C4 give frozen three-value descent. Lemma C5 bounds both parameter changes by summable sequences. Lemma C6 combines these estimates into a finite master budget. Lemma C8.3 selects a single tail index κ for every term in that budget.
At this index, Lemma C8.1 and the bounds

$$\|\Delta \boldsymbol{x}^{\kappa+1}\| = O(K^{-1/2}), \qquad \|\boldsymbol{z}^{\kappa} - \boldsymbol{x}^{\kappa+1}\| = O(K^{-1/3})$$

give the stationarity estimate. Lemmas C7.2 and C8.2, together with

$$\|\boldsymbol{\lambda}^{\kappa+1} - \boldsymbol{\lambda}^{\kappa}\| = O(K^{-1/2}), \qquad r_{\kappa} = O(K^{-1/3}),$$

give the feasibility estimate. The subgradient relation is exact by Lemma C8.1. Therefore the split-KKT residual is $O(K^{-1/3})$. Choosing K so that $CK^{-1/3} \le \varepsilon$ gives

$$K = O(\varepsilon^{-3}).$$

Each iteration uses one gradient evaluation at $\boldsymbol{x}^{k}$, one explicit dual gradient, two evaluations of $\mathrm{prox}_{h/\rho}$, and two center averages. Hence the gradient and proximal-oracle complexities are both $O(\varepsilon^{-3})$. □